Relief load basis and PSV-400 sizing

D-400 flash drum system · API 521 causes of overpressure, API 520 sizing, model relief runs, flare loads

Document no. FD-V1-PE-RLB-001

Releases the HOLD of FD-400-PID-001 note 3: every credible cause of overpressure of the D-400 system with its relieving conditions and load, the governing case, the PSV-400 orifice, the inlet-loss and back-pressure checks and the flare loads. The overpressure scenarios were also run on copies of the simulator with the SIS trips disabled.

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Note. Numbers come from the simulator model (design case and steady model runs) and the stated assumptions; generated for teaching, not for construction. Engineering queries raised by this document are listed in it and in FD-V1-PM-DOC-001.

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DOCUMENT NO.	REV	STATUS	ORIGINATOR	DATE
FD-V1-PE-RLB-001	0	Issued for teaching	Apinan Soottitantawat	28-OCT-26

1 · Protected system and basis

Device	PSV-400, conventional spring-loaded valve on the D-400 vapour line, to the closed flare header
Protected system	D-400, the let-down outlet from FCV-400, the vapour line to PCV-420, the bottoms line to LCV-400
Set pressure / MAWP	450 kPa g / 450 kPa g stamped (calculated 684 kPa g)
Relieving pressure	596.3 kPa a (10 % overpressure, process causes); 645.8 kPa a (21 %, fire)
Back-pressure	superimposed 10 kPa g (assumed); built-up 6.2 kPa in the 6" tail pipe
Fluid properties	model thermodynamics (Antoine + Raoult, Watson latent heat); ideal gas Z = 1 (conservative), k = 1.08 (typical), Kd 0.975 vapour / 0.65 liquid
Credit for the SIS	none in the relief load (API 521); the trips (I-1) act first in every model run

2 · Relief load basis

Case	Cause of overpressure	Basis	P1 kPa a	Phase	T °C	M	Load kg/h	Area mm²	Status	How the load was found	Model run
C1	Blocked vapour outlet (PCV-420 or HV-421 closed), hot feed at 74 °C	API 521 blocked outlet: vapour generated at the relieving pressure	596.3	—	—	—	0	0.0	credible	Feed bubble pressure at 74 °C is 359 kPa g < relieving 495 kPa g: no vapour forms, the pressure stalls near 350 kPa g (model). PSHH-426 trips at 250 kPa g after 966 s.	blk74
C2	Blocked vapour outlet with the hot feed at 90 °C (TIC-401 set point / TT-401 error)	Flash of the full feed at the relieving pressure	596.3	V	86.4	62.69	296	50.0	credible	Bubble pressure 559 kPa g > relieving: 3.41 mol% flashes at 596 kPa a. Model run: PSV-400 lifts after 1607 s, up to 1442 kg/h at ≈ 445 kPa g (the model's PSV opens below set).	blk90
C3	Blocked vapour outlet + TCV-401 fully to E-400 (two failures; bounding case of the Design tab)	Flash at the relieving pressure of the hottest feed E-400 can make	596.3	V	89.2	63.38	1,050	175.0	two failures	Hot feed 101.9 °C (model, TCV-401 100 %), bubble pressure 745 kPa g; 11.97 mol% flashes. Model: lifts after 996 s, 1233 kg/h at 442 kPa g.	blkTCV
C4	Control-valve failure: FCV-400 wide open (also with the globe bypass HV-403 open)	API 521 inlet control-valve failure, credit for the normal vapour outlet	596.3	—	—	—	0	0.0	credible	Feed rises to 28.4 m³/h (31.6 with HV-403); PCV-420 opens to 55.6 % and the pressure peaks at 107.4 kPa g (model). PCV-420 at 100 % passes 13806 kg/h. HV-402 open changes nothing (it bypasses the block valve, FIC-400 still controls).	fcv
C5	Nitrogen regulator PCV-430 fails wide open	Gas inflow at the relieving pressure, no credit for PCV-420 (conservative)	596.3	N_2	28.0	28.01	154	32.0	credible	700 kPa g header through C_v 1.5 (assumed regulator trim) into 596 kPa a: 154 kg/h. The HAZOP asks for a restriction orifice after PCV-430 (R-15).	—
C6	External fire (pool fire under D-400)	API 521: $Q = 43.2 \cdot F \cdot A_{wet}^{\ 0.82}$ kW, adequate drainage and fire fighting, F = 1 (bare)	645.8	V	96.7	64.48	4,247	656.0	GOVERNING	A_{wet} 11.82 m² (bottom head + shell to NLL 1.8 m above BT, all below 7.6 m); Q 327 kW; liquid boils at 96.7 °C at 646 kPa a, λ 277.6 kJ/kg → W = Q/λ.	—
C7	Thermal expansion of the blocked-in E-400 tube side (XV-408 and FCV-400 closed, hot oil still flowing)	API 521 hydraulic expansion: $q = \alpha \cdot \varphi / (\rho \cdot c_p)$	—	L	74.0	—	1,026	13.6	other device	$\varphi = UA\,(T_{oil} - T_{feed})$ = 412 kW, α 0.0016 1/K (assumed): 1.79 m³/h. Not PSV-400 (outside the D-400 system): needs a thermal relief valve on E-400 (none on the P&ID). Above 111 °C the trapped feed boils at the TSV set.	—
C8	Gas blow-by: level lost, LCV-400 wide open (downstream receiver)	Vapour through LCV-400 at 100 % into the 40 kPa g receiver	—	V	49.2	62.86	2,274	—	other device	Drum vapour through the model's LCV-400 capacity: 2274 kg/h at 102.7 kPa g, 7837 kg/h with the drum at the PSV set (11211 kg/h with HV-413 also open). Relief of the downstream receiver, outside this unit; I-2 (LSLL-417 → XV-418) is the protection.	—
C9	Liquid overfill (blocked liquid outlet, LSHH-416 fails) with the vapour outlet also closed	Liquid relief, API 520 liquid sizing, K_d 0.65 (two failures)	596.3	L	74.0	—	9,642	175.0	two failures	Feed through the supply curve against the relieving pressure: 16.9 m³/h of hot liquid. With the vapour outlet open the model carries the liquid into the vapour line (run: level 120 %, AI-420 35.5 wt%) and the pressure does not rise: overfill is an SIS case (I-1, LSHH-416), not a PSV case.	liq
C10	E-400 tube rupture	API 521 10/13 rule	—	—	—	—	0	0.0	not credible	High-pressure side: hot oil (shell) design 1000 kPa g; low side: feed (tubes) 820 kPa g = 82 % ≥ 10/13 (77 %): not a relief case. Oil would leak into the feed (quality).	—
C11	Utility failures (power, instrument air, hot oil)	API 521 utility failure	—	—	—	—	0	0.0	credible	No motors in the unit; instrument air failure sends FCV-400 and LCV-400 closed, TCV-401 to bypass and PCV-420 open (depressures to the header). Hot-oil failure removes the heat. None raises the pressure.	—

3 · Governing case: external fire

Step	Relation	Result
Wetted area	bottom head 1.084 D^2 + π D h to NLL (1.8 m above BT, BT 3 m above grade, all below 7.6 m)	11.82 m²
Heat input	$Q = 43.2\ F\ A^{0.82}$ kW, F = 1 (bare), adequate drainage and fire fighting	327.4 kW (537.4 kW without drainage)
Relieving temperature	bubble point of the drum liquid at 645.8 kPa a	96.69 °C, vapour y = 0.7732, M = 64.48
Latent heat	$\lambda = (y\ \Delta H_{C4} + (1 - y)\ \Delta H_{C6})/M$ (model)	277.6 kJ/kg
Load	W = Q/λ	4,247 kg/h (6,970 kg/h without drainage)
Area	A = W/(C Kd P1 Kb Kc) √(TZ/M), C = 0.02464	655.5 mm² → J (830 mm², 2J3)
Design tab short-cut	same method, λ from the liquid	3943 kg/h, 604 mm² → J

4 · PSV-400 selection and installation checks

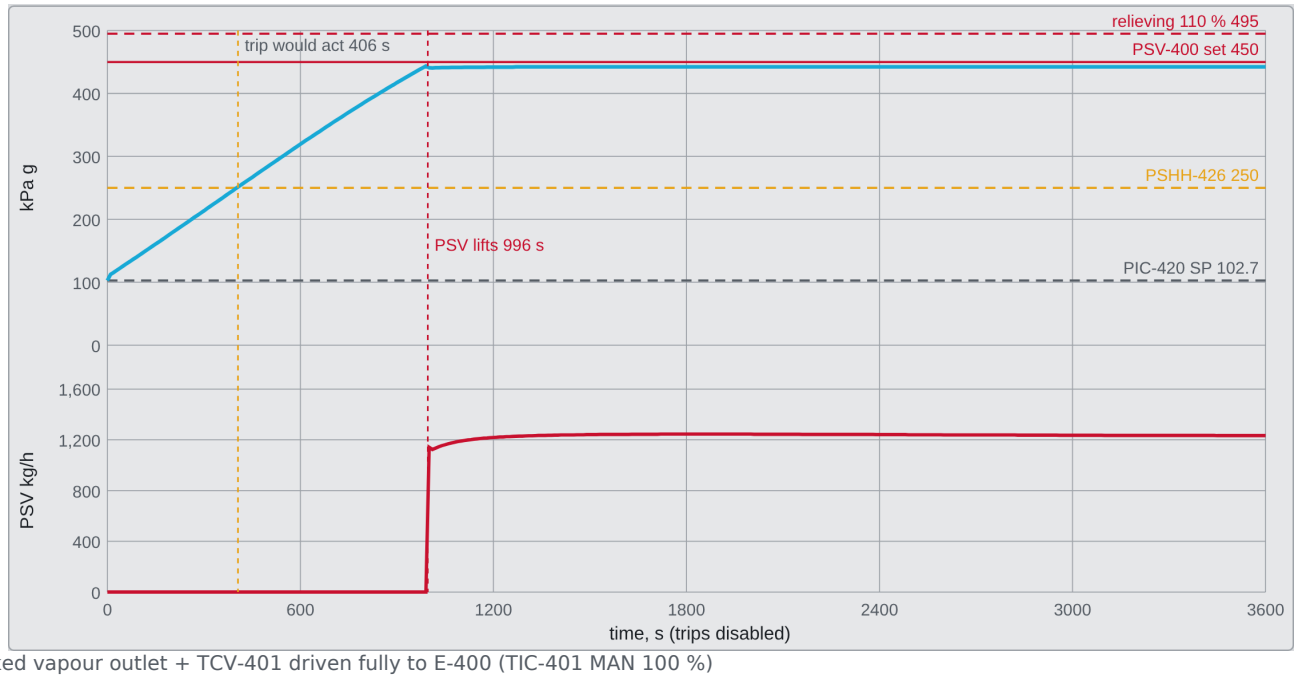
Selected	2J3: orifice J, 830 mm², conventional, closed bonnet; inlet line 3"-PV-4023-A1A, tail pipe 6"-FL-4041-F1A
Capacity	5,377 kg/h in the fire case (79 % used); 4,973 kg/h of blocked-outlet vapour at 110 %
Inlet loss (3 % rule)	6.81 kPa = 1.51 % of set at the rated flow; 4" N2 nozzle and 1.0 m of 4"-PV-4021-A1A (K 0.5 entrance), 3" branch 0.8 m with tee and block valve (K 1.2), all at the rated capacity of a J orifice in the fire case
Built-up back-pressure (10 % rule)	6.2 kPa = 1.4 % of set with a 6" tail pipe, 45 m equivalent (assumed route)

Tail pipe	P at outlet kPa a	Built-up kPa	% of set	Mach	≤ 10 %
3"	260.4	149.1	33.1	0.253	✗
4"	158.1	46.8	10.4	0.241	✗
6"	117.5	6.2	1.4	0.143	✓
8"	112.8	1.5	0.3	0.086	✓

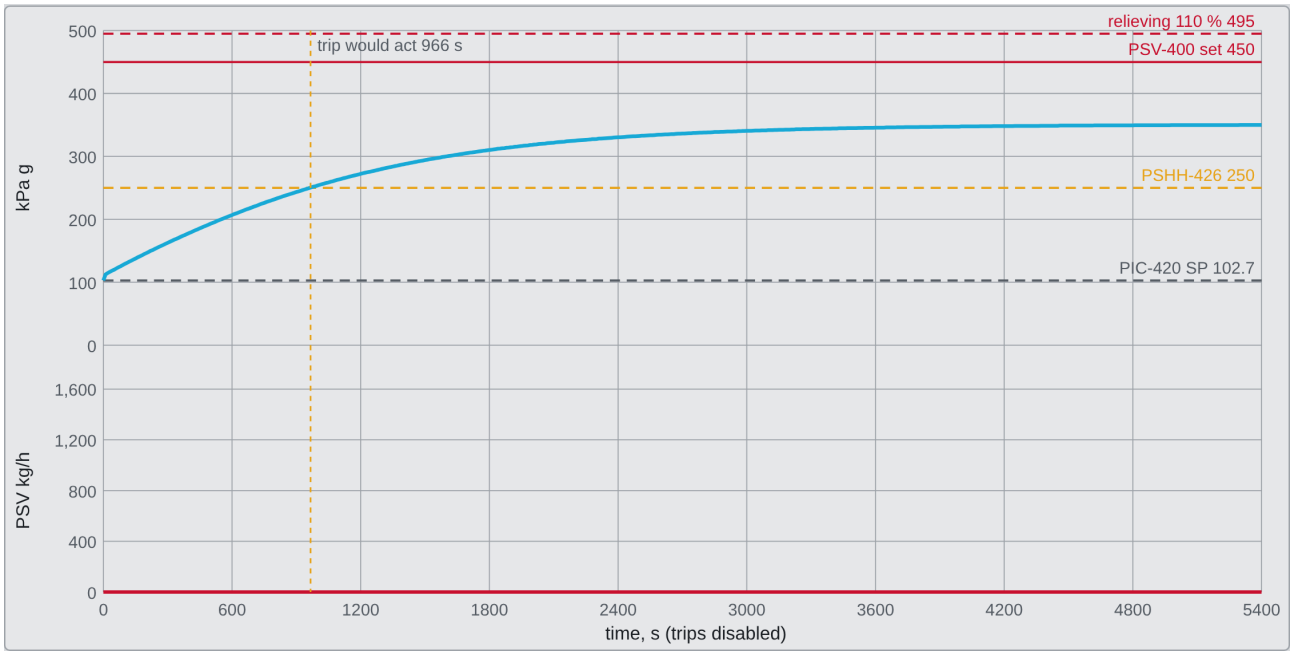
5 · Model relief runs

Copies of the simulator from the design state, one failure each, SIS trips disabled (K.Lshh, K.Pshh, K.Lsll out of reach). The model PSV pops at 445 kPa g and passes 0.02 kg/s per kPa above 425 kPa g (a modulating simplification; query ER-04).

Run	Scenario	s	Peak kPa g (at s)	PSHH-426 would trip at s	PSV lifts at s	Max relief kg/h
blk74	Blocked vapour outlet (HV-421 closed), hot feed held at 74 °C	5,400.0	349.9 (5400)	966.0	no lift	0
blk90	Blocked vapour outlet with the hot feed at 90 °C (TIC-401 set point raised)	3,600.0	445.0 (1606)	545.0	1,607.0	1,442
blkTCV	Blocked vapour outlet + TCV-401 driven fully to E-400 (TIC-401 MAN 100 %)	3,600.0	445.1 (995)	406.0	996.0	1,444
tcv	TCV-401 driven fully to E-400 (outlet open)	1,800.0	109.5 (160)	—	no lift	0
fcv	FCV-400 wide open (FIC-400 MAN 100 %)	1,800.0	107.4 (8)	—	no lift	0
fcv403	FCV-400 wide open and the globe bypass HV-403 wide open	1,800.0	108.6 (7)	—	no lift	0
liq	Blocked liquid outlet (HV-411 closed): overfill	1,800.0	103.1 (126)	—	no lift	0



Blocked vapour outlet + TCV-401 driven fully to E-400 (TIC-401 MAN 100 %)

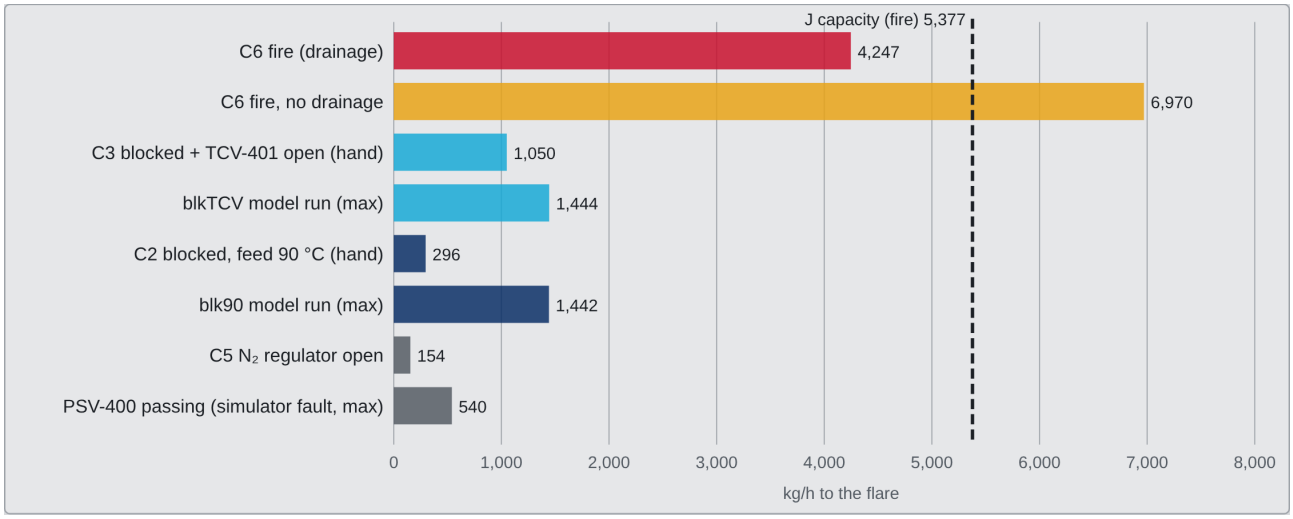


Blocked vapour outlet (HV-421 closed), hot feed held at 74 °C

6 · Other devices and cases

Case	Load	Device	Note
Thermal expansion, E-400 tube side blocked in	1.794 m³/h (412.3 kW, α 0.0016 1/K)	TSV-400 proposed, D (1D2), set 820 kPa g	not on the P&ID; boils above 110.6 °C (ER-02)
Gas blow-by through LCV-400 wide open	2,274 kg/h (7,837 at the PSV set, 11,211 with HV-413)	downstream receiver relief (other unit)	I-2 LSLL-417 → XV-418 protects (ER-03)
Nitrogen regulator PCV-430 wide open	154 kg/h N_2 (Cv 1.5 assumed)	PSV-400	3.1 % of the J capacity; RO after PCV-430 (HAZOP R-15, ER-06)
Overfill with the vapour outlet closed	9,642 kg/h liquid	PSV-400 as a liquid valve (175 mm²)	two failures; the model sends the liquid to the vapour line (ER-05)
E-400 tube rupture	none	—	low side 820 ≥ 10/13 × 1000 kPa g

7 · Flare and vent loads, depressuring



API 521 asks for a depressuring system mainly for vessels operating above about 1700 kPa g or exposed to fire with a thin wall above the liquid. D-400 runs at about 0.1 MPa g with a 450 kPa g design: no automatic depressuring valve is proposed. The drum can be depressured by hand to the vapour header through PCV-420 / HV-423 and, for maintenance, to the flare through a vent (query on the missing vent nozzle).

8 · Engineering queries

No.	Subject	Query	Why it matters	Proposed answer	Status
ER-01	PSV-400 orifice on HOLD	P&ID note 3 holds the relief cases and size. The governing case is the external fire (4247 kg/h, 656 mm²) → orifice J (830 mm², 2J3); the Design tab also selects J.	The valve cannot be bought without the orifice and the inlet / outlet sizes.	Release the HOLD: PSV-400 2J3 (J, 830 mm²), conventional, set 450 kPa g, inlet 3"-PV-4023-A1A, tail pipe 6"-FL-4041-F1A.	OPEN
ER-02	No thermal relief on the E-400 tube side	XV-408 and FCV-400 both fail closed (I-1, air failure) and can trap the feed in E-400 while hot oil still flows (XV-409 may stay open on a non-SIS closure). Expansion 1.79 m³/h at the start, boiling above 111 °C.	Blocked-in liquid heated by 180 °C oil reaches the tube design pressure in seconds.	Add TSV-400 (¾" × 1", D orifice, set 820 kPa g) on the E-400 tube outlet, discharging to D-400.	OPEN

No.	Subject	Query	Why it matters	Proposed answer	Status
ER-03	Gas blow-by into the downstream receiver	With the level lost and LCV-400 wide open, 2274 kg/h of drum vapour flows to the 40 kPa g receiver (7837 kg/h at the PSV set, 11211 kg/h with HV-413 open).	The receiver relief belongs to the downstream unit; this unit relies on I-2 (LSLL-417 → XV-418).	Send the blow-by rate to the downstream unit for its relief study; confirm the SIL of I-2 in the SRS.	OPEN
ER-04	Model PSV characteristic	The model's PSV-400 pops at 445 kPa g and passes 0.02 kg/s per kPa above 425 kPa g: 5040 kg/h at 495 kPa g. A J orifice passes 4973 kg/h of blocked-outlet vapour at 495 kPa g and 5377 kg/h in the fire case at 545 kPa g.	The trainee sees PSV-400 lift a little below set and modulate; a real conventional valve pops at set and runs full open.	Keep the model (teaching) and state the difference in the Relief tab; the capacity is of the right order for a J valve.	OPEN
ER-05	Overfill goes to the vapour header	In the model a blocked liquid outlet fills D-400 and carries liquid into the vapour line (AI-420 falls to 35.5 wt%); the pressure does not rise and PSV-400 never sees liquid.	The fuel-gas / vapour header must not receive liquid; I-1 (LSHH-416) is the only barrier.	Confirm a knock-out drum on the vapour header downstream; keep LSHH-416 as the safeguard (HAZOP).	OPEN
ER-06	PCV-430 capacity and restriction orifice	The regulator size is not on the P&ID; with C_v 1.5 (assumed) a wide-open PCV-430 passes 154 kg/h N_2 at the relieving pressure (orifice J holds it easily). The HAZOP (R-15) asks for a restriction orifice.	The N_2 case must stay below the PSV capacity whatever regulator is bought.	State the regulator C_v and size the RO so that the N_2 flow at 700 kPa g ≤ 10 % of the PSV capacity.	OPEN
ER-07	Flare superimposed back-pressure assumed	The built-up back-pressure (6.2 kPa, 1.4 % of set in 6"; 10.4 % in 4") was calculated with 10 kPa g superimposed in the flare header (assumed) and a 45 m equivalent tail pipe (assumed route).	A conventional valve tolerates ≤ 10 % built-up; a higher header pressure would call for a balanced bellows valve.	Obtain the flare-header back-pressure from the flare study.	OPEN